

1:

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

A particle P moves along the x -axis.

At time $t = 0$, P is at rest at the origin O .

At time t seconds, $t \geq 0$, the acceleration of P , in the positive x direction, is

$$k(3 - t^2) \text{ m s}^{-2}$$

where k is a positive constant.

Given that when $t = 1.5$, the speed of P is 13.5 m s^{-1}

(a) show that $k = 4$

(3)

The displacement of P from O is s metres in the positive x direction.

(b) Show that

$$s = \frac{1}{3} t^2 (18 - t^2)$$

(2)

(c) Find the total distance that P moves before it returns to O .

(4)

2:

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

[In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P of mass 3 kg is moving on a smooth horizontal plane.

At time t seconds, where $1 \leq t \leq 4$, the position vector, $\mathbf{r}$ metres, of P relative to a fixed origin is given by $\mathbf{r} = 4t^{\frac{1}{2}} \mathbf{i} - 3t \mathbf{j}$.

At the instant when $t = 4$, P receives an impulse $(6\mathbf{i} - 3\mathbf{j}) \text{ N s}$.

Find the speed of P immediately after the impulse is received.

- 3: [The distance of the centre of mass of a semicircular arc of radius r is $\frac{2r}{\pi}$ from its centre.]

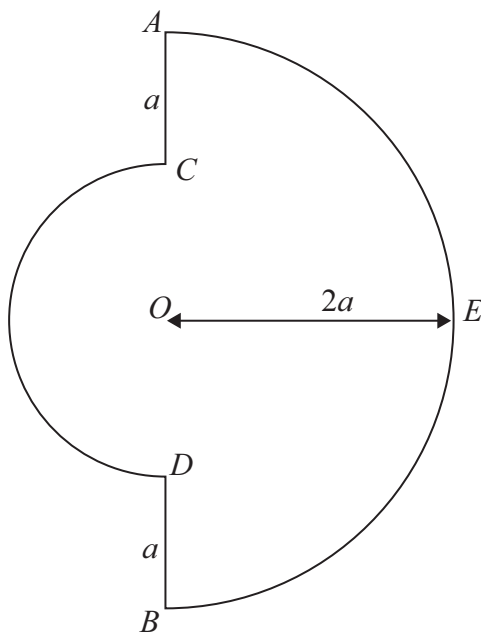


Figure 1

A uniform rigid wire **framework**, shown in Figure 1, consists of two wire semicircular arcs AB and CD , both with centre O , joined by two straight pieces of wire, AC and DB .

Arc AB has radius $2a$, arc CD has radius a , $AC = DB = a$, and $ACDB$ is a straight line.

The framework lies in a single plane.

- (a) Show that the centre of mass of the framework is a distance $\frac{6a}{(3\pi + 2)}$ from AB .

(5)

The point E is on the arc AB and the angle AOE is a right angle.

The framework is freely suspended from A .

The framework hangs at rest in a vertical plane with AE making an angle θ with the downward vertical.

- (b) Find the exact value of $\tan \theta$, simplifying your answer.

(5)

4: A rough plane is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{1}{7}$

A car of mass 750 kg moves up a line of greatest slope of the plane. The total resistance to the motion of the car from non-gravitational forces is 200 N. The engine of the car is working at a constant rate of 20 kW.

(a) Find the constant speed, in ms^{-1} , at which the car could move up the slope.

(4)

(b) Find the acceleration of the car at the instant when it is moving up the slope at 10ms^{-1}

(4)

At this instant, the car is at the point A and the engine of the car is switched off.

The total resistance to the motion of the car from non-gravitational forces remains at 200 N.

Using the work-energy principle,

(c) find how far up the slope from A the car would travel before coming to rest.

(4)

- 5: A particle P of mass $2m$ and a particle Q of mass m are at rest on a smooth horizontal plane. Particle P is projected with speed u along the plane towards Q and the particles collide. The coefficient of restitution between the particles is e , where $e > 0$

(a) Show that the speed of Q after the collision is $\frac{2(1+e)u}{3}$ (5)

The total loss of kinetic energy due to the collision is $k(1 - e^2)mu^2$, where k is a constant.

(b) Find the value of k . (5)

The magnitude of the impulse of Q on P in the collision is $\frac{8mu}{9}$

(c) Find the value of e . (2)

After the collision, Q goes on to hit a vertical wall which is fixed at right angles to the direction of motion of Q . The coefficient of restitution between Q and the wall is f .

After Q has hit the wall, Q has the same speed as P .

(d) Find the value of f . (3)

6:

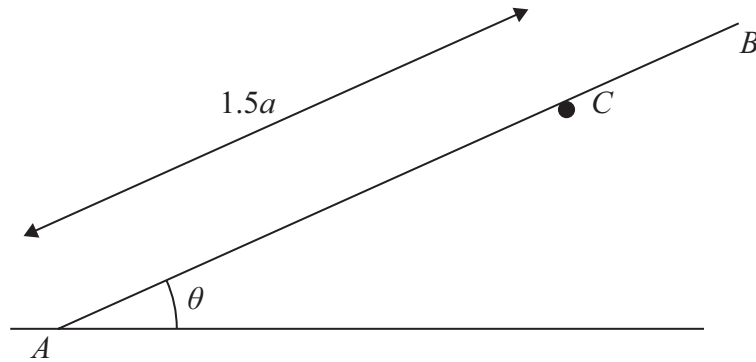


Figure 2

A uniform rod AB , of mass m and length $2a$, has its end A on a rough horizontal floor.

A smooth horizontal rail is fixed above the floor.

The rod rests in equilibrium against the rail at the point C , where $AC = 1.5a$.

The vertical plane containing the rod is at right angles to the rail.

The rod is inclined to the floor at an angle θ , as shown in Figure 2.

The magnitude of the normal reaction exerted on the rod at C is S .

(a) Show that $S = \frac{2mg \cos \theta}{3}$

(3)

The magnitude of the vertical component of the force exerted on the rod at A is V .

(b) Show that $V = \frac{mg}{3} (3 - 2 \cos^2 \theta)$

(3)

Given that the rod is in limiting equilibrium and the coefficient of friction between the rod and the floor is $\frac{4}{7}$

(c) show that

$$p \tan^2 \theta + q \tan \theta + r = 0$$

where p , q and r are integers to be found.

(6)

7: A small stone is projected from a point O on horizontal ground.

The stone is projected with speed 20 m s^{-1} at an angle α to the ground.

When the stone is 11 m above the ground

- the vertical velocity component of the stone has magnitude $V \text{ m s}^{-1}$
- the horizontal velocity component of the stone has magnitude $2V \text{ m s}^{-1}$

In a model of the motion

- the stone is modelled as a particle moving freely under gravity
- **the acceleration due to gravity is modelled as being 10 m s^{-2}**

Using the model and the principle of conservation of mechanical energy,

(a) show that $V = 6$

(4)

Figure 3 shows a plane fixed to the ground at an angle of 60°

The stone hits this plane at the point A .

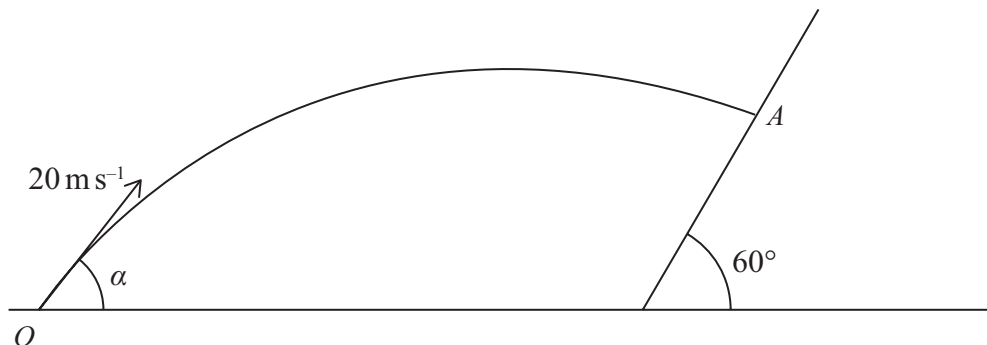


Figure 3

The direction of the motion of the stone is perpendicular to the plane at the instant the stone hits A .

Using the model,

(b) find the height of A above the ground, giving your answer to 2 significant figures.

(6)